Heat-Risk Viewer: PCA Pillar Analysis

Method as implemented, with per-component loadings

San Diego County Heat-Risk Viewer

May 21, 2026

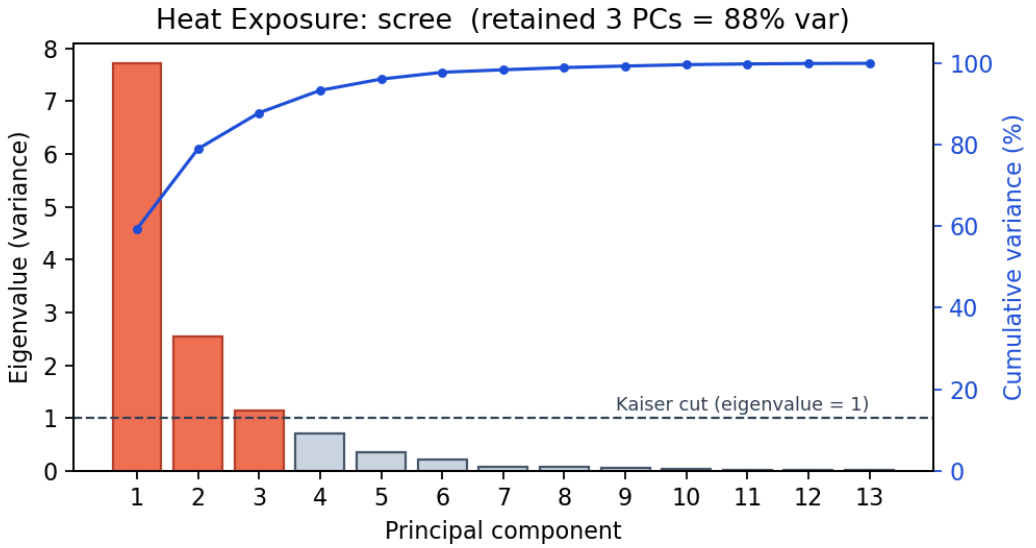
How each pillar score is built

Three pillars — **Heat Exposure**, **Population Sensitivity**, **Adaptive Capacity** — each reduced to one tract-level score by PCA on the variables the user enables. Computed client-side (Jacobi eigendecomposition).

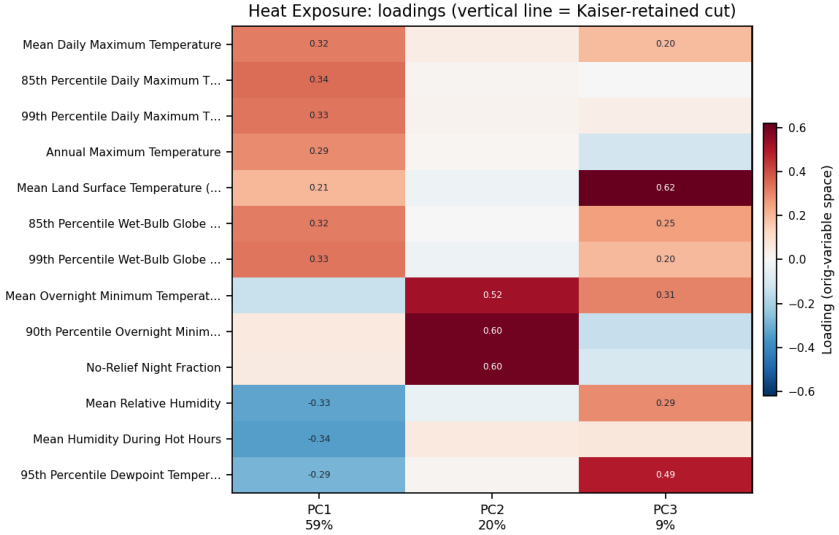
1. **Standardize**: z-score each variable across all 735 tracts.
2. **Polarity-align**: flip variables so higher always means higher risk.
3. **Decompose**: eigendecomposition of the correlation matrix.
4. **Retain**: keep PCs with eigenvalue > 1 (Kaiser rule), at least 1.
5. **Composite**: variance-weighted sum of retained PC scores $w_i = \sum_{k \le n_{\text{ret}}} \frac{\lambda_k}{\sum \lambda} s_{ik}$, sign-aligned to a reference variable.
6. **Rank**: percentile-rank w across the county (0–100).

Composite risk = mean of the three pillar percentile scores (Adaptive Capacity inverted). Reference variables: mean T_{max} (exposure), high blood pressure (sensitivity), income (adaptive).

Heat Exposure — variance structure



Heat Exposure — loadings on each PC

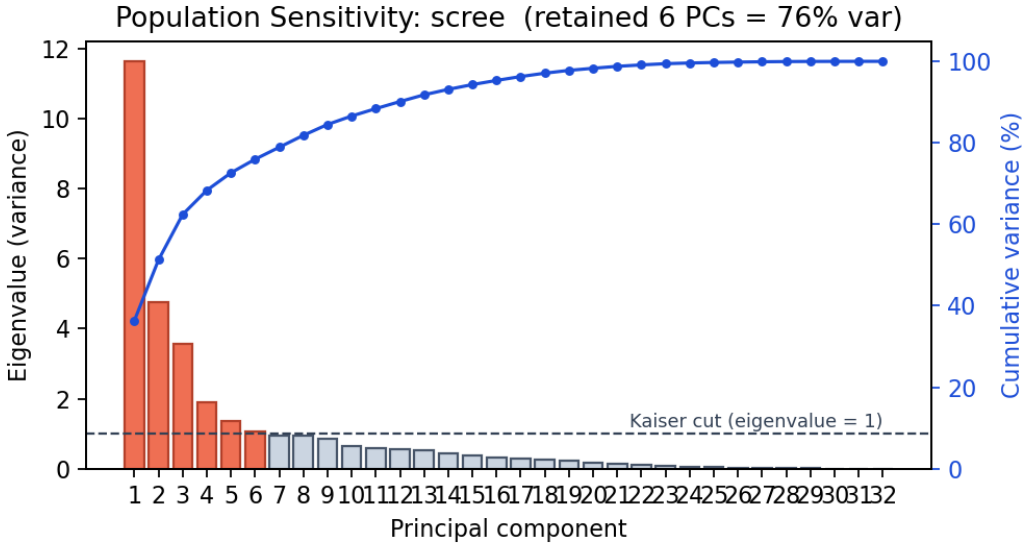


Heat Exposure — strongest loadings per component

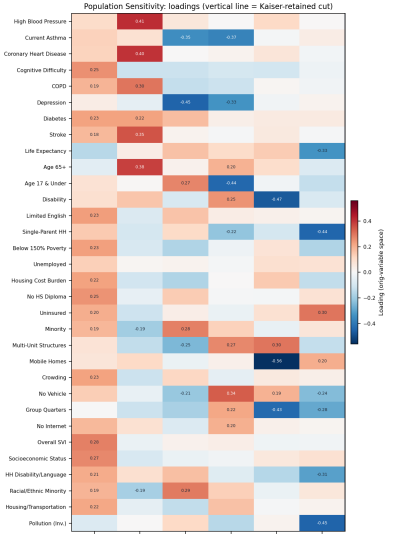
PC1 (59%)	**PC2** (20%)	**PC3** (9%)
85th Percentile Daily... (+0.34)	No-Relief Night Fract... (+0.60)	Mean Land Surface Tem... (+0.62)
Mean Humidity During ... (-0.34)	90th Percentile Overn... (+0.60)	95th Percentile Dewpo... (+0.49)
99th Percentile Wet-B... (+0.33)	Mean Overnight Minimu... (+0.52)	Mean Overnight Minimu... (+0.31)
99th Percentile Daily... (+0.33)	Mean Humidity During ... (+0.06)	Mean Relative Humidity (+0.29)
Mean Relative Humidity (-0.33)	Mean Daily Maximum Te... (+0.05)	85th Percentile Wet-B... (+0.25)
85th Percentile Wet-B... (+0.32)	Mean Relative Humidity (-0.04)	99th Percentile Wet-B... (+0.20)

Each cell: variable (signed loading in original-variable space). Positive = higher value of that variable pushes the pattern toward higher risk.

Population Sensitivity — variance structure



Population Sensitivity — loadings on each PC

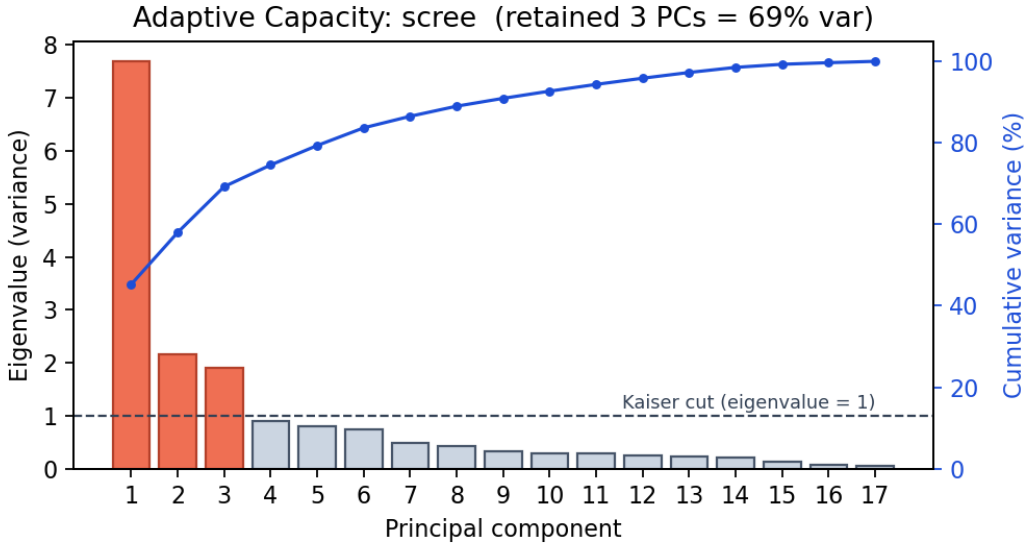


Population Sensitivity — strongest loadings per component

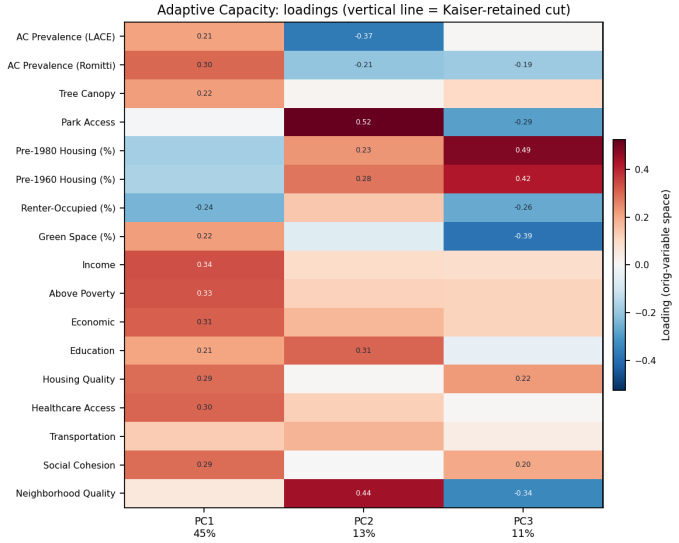
PC1 (36%)	**PC2** (15%)	**PC3** (11%)
Overall SVI (+0.28)	High Blood Pressure (+0.41)	Depression (-0.45)
Socioeconomic Status (+0.27)	Coronary Heart Disease (+0.40)	Current Asthma (-0.35)
Cognitive Difficulty (+0.25)	Age 65+ (+0.38)	Racial/Ethnic Minority (+0.29)
No HS Diploma (+0.25)	Stroke (+0.35)	Minority (+0.28)
Limited English (+0.23)	COPD (+0.30)	Age 17 & Under (+0.27)
Below 150% Poverty (+0.23)	Diabetes (+0.22)	Multi-Unit Structures (-0.25)

Each cell: variable (signed loading in original-variable space). Positive = higher value of that variable pushes the pattern toward higher risk.

Adaptive Capacity — variance structure



Adaptive Capacity — loadings on each PC



Adaptive Capacity — strongest loadings per component

PC1 (45%)	**PC2** (13%)	**PC3** (11%)
Income (+0.34)	Park Access (+0.52)	Pre-1980 Housing (%) (+0.49)
Above Poverty (+0.33)	Neighborhood Quality (+0.44)	Pre-1960 Housing (%) (+0.42)
Economic (+0.31)	AC Prevalence (LACE) (-0.37)	Green Space (%) (-0.39)
Healthcare Access (+0.30)	Education (+0.31)	Neighborhood Quality (-0.34)
AC Prevalence (Romitt... (+0.30)	Pre-1960 Housing (%) (+0.28)	Park Access (-0.29)
Housing Quality (+0.29)	Pre-1980 Housing (%) (+0.23)	Renter-Occupied (%) (-0.26)

Each cell: variable (signed loading in original-variable space). Positive = higher value of that variable pushes the pattern toward higher risk.

What the composite actually is

The pillar score collapses the retained PCs back to one number — a linear weighted combination of the inputs. Because a PC's spread scales as $\sqrt{\lambda}$ *and* it is weighted by λ, the contribution of each component scales as $\lambda^{1.5}$: PC1 dominates.

Pillar	PC1 var	Retained	Spearman of score vs.	
			equal-wt avg	PC1 alone
Heat Exposure	59%	3 (88%)	0.68	0.99
Population Sensitivity	36%	6 (76%)	0.87	0.93
Adaptive Capacity	45%	3 (69%)	0.96	0.98

PCA's value here is **redundancy-aware weighting** ($\approx$ PC1): the more correlated the inputs (e.g. exposure's many temperature measures), the more the score departs from a plain average. The retained-count / cumulative-variance figures describe the PC *set*, not how much the final score uses — that is effectively PC1.